

Workshop on Novel Input Devices and Interaction Techniques 2025

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1 INTRODUCTION

Virtual Reality (VR), augmented reality (AR), and mixed/extended reality (MR/XR) have become a mainstream technology. Recent advances in VR display technologies have led to high-resolution, ergonomic, and – critically – low-cost head-mounted displays (HMDs). Advances in commercial input devices and interaction techniques have arguably not kept pace with these advances in display systems. For instance, most HMDs include tracked input devices, but these “controllers” are still fairly similar to the earliest examples of 3D controllers used in the VR systems of the 1980s. Interaction in commercial VR systems has similarly lagged; despite many advances in 3D interaction in the past three decades of VR research, interaction in commercial systems largely relies on classical techniques like the virtual hand or ray-casting.

Moreover, these are challenging times for VR, AR, and MR/XR researchers, with the technology more readily available to the general public than ever before. Yet, there seems to be a lack of interest in technology that has caused people to see it only as a video game console or an enjoyable but unnecessary experience. To overcome these challenges, the **6th workshop on Novel Input Devices and Interaction Techniques (NIDIT’25)** aims to drive innovation in hardware and software for 3D interactive systems. **NIDIT’25** provides substantial value as a place that attracts contributions from researchers and practitioners desiring to shape the future of input devices. The accepted papers covered new input devices and/or interaction techniques and/or 3D user interfaces (3DUI) in 3D/spatial situations, including VR, AR, and MR/XR. Moreover, **NIDIT’25** has a practical hands-on component, through presenters also demonstrating their research, helping participants understand the suggested 3DUI.

NIDIT’25 welcomes a wide-ranging audience, including researchers and academics specializing in computer science, HCI, and cognitive science, as well as domain professionals like VR/AR developers, hardware and software manufacturers, designers, artists, educators, entrepreneurs, and UX/UI designers. Researchers and academics value seeing the latest advancements and methodologies, while developers and manufacturers can explore practical applications and innovative technologies. Designers and artists benefit from new creative possibilities, educators can enhance educational content, and entrepreneurs will gain insights into market demands. Additionally, UX/UI designers gain valuable knowledge about user-friendly interaction techniques in immersive

environments. The workshop’s diverse audience also fosters collaboration, knowledge exchange, and innovation in the ever-evolving field of immersive technologies.

2 THE WORKSHOP

Robert J. Teather and Adalberto L. Simeone, with the help of Francisco R. Ortega, organized the first **NIDIT’25** workshop at IEEE 2019 in Osaka, Japan. Since then, the workshop has received high-quality papers on all topics, with all papers published at the IEEE Xplore library. Over the years, attendance numbers have been high, and the workshop is usually a full house, with attendants usually having to listen to talks standing.

The 6th **NIDIT’25** was held on March 9th, 2025, as part of the 32nd IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR 2025). The workshop had nine submissions and accepted seven papers, with an acceptance rate of 77.8%. At least two workshop co-chairs, experts in virtual, augmented, and mixed reality and 3D user interfaces, reviewed each submission.

Participants submitted 4- and 6-page papers addressing designing, developing, and evaluating new input devices and interaction techniques for immersive environments. This includes hardware and software solutions, as well as user studies. Authors were also strongly encouraged (although not required) to present a live demonstration of their work at the workshop. Workshop papers include the following topics:

- Form factors and ergonomics of input devices
- Hardware design and prototyping
- Mapping of input to varying degrees of freedom
- Haptic/tactile feedback
- Novel input devices
- Repurposing of existing devices (e.g., smartphones, tablets) for immersive contexts
- Tracked passive or custom props
- Novel interaction techniques supported by custom devices
- User studies evaluating the above topics

3 ACCEPTED PAPERS

Here is a list of all accepted papers and authors of the 6th workshop on Novel Input Devices and Interaction Techniques (**NIDIT’25**):

- **An Approach for Effective CPR Trainings in Virtual Reality with Multimodal Feedback:** Suyash Aditya Lal, Chiranjay Chattopadhyay and Dr Rahul Kumar
- **MUHI - A Multi-Use Haptic Interface that enables Haptic Interactive Distractors for Redirected Walking:** Fabian Rücker, Torben Storch and Eike Langbehn

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- **Logitech MX Ink: Extending the Meta Quest Platform with a 6DoF Stylus:** Aidan Kehoe, Dr. Mario Gutierrez, Yena Ahn and Vadim Kogan
- **Interactive Machine Learning for Movement Interaction in VR:** Tom Lawrence, Tianyuan Zhang, Clarice Hilton and Prof Marco Fyfe Pietro Gillies
- **Does Perspective Matter? Understanding the Role of Viewpoints on User Performance in 3D Sketching:** Jialin Zhang and Mayra Donaji Barrera Machuca
- **A Novel Bare-Handed Manipulation Technique for Distant Objects in Virtual Reality:** Di Zhao and Wolfgang Stuerzlinger
- **A Haptic Device for Tennis Simulation: Dual-Flywheel System for Rendering Virtual Impact:** Allyson E. Chen, Xuan Gedney and Jasmine Roberts